Weinberg Institute for Theoretical Physics
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Jonathan Lozano-Mayo

PhD Candidate in Theoretical Physics

Research Interests

Condensed matter physics · Many-particle physics · Quantum phenomena
Tensor networks for strongly correlated systems · Non-perturbative phenomena
Topological and non-topological solitons

Education

2023–present **Ph.D. in Physics**, *The University of Texas at Austin*, Weinberg Institute for Theoretical Physics
Advised by Antonios Alvertis.

2016–2021 **B.Sc. in Physics**, *National Autonomous University of Mexico (UNAM)*, Faculty of Sciences
Advised by Manuel Torres Labansat.
Thesis: Spontaneous symmetry breaking and extended field configurations in a scalar theory subject to a potential with two families of vacuum states. HONORIFIC MENTION

Publications

Refereed journal articles

2026 **J. Lozano-Mayo**, *Quantum vacuum inversion and multi-kink unbinding in the non-degenerate double sine-Gordon model* [arXiv:2609.24861]. Accepted at JHEP.

2026 **J. Lozano-Mayo**, *Generalized virial identities: radial constraints for solitons, instantons, and bounces*, J. High Energ. Phys. **04** (2026) 058 [arXiv:2512.23548].

2026 **J. Lozano-Mayo** and M. Torres-Labansat, *Multi-kinks and composite oscillons in a commensurable and non-degenerate double sine-Gordon model*, J. High Energ. Phys. **09** (2026) 218 [arXiv:2407.09682].

2021 **J. Lozano-Mayo** and M. Torres-Labansat, *Kink solutions in a generalized scalar ϕ^4_G field model*, J. Phys. Commun. **5** (2021) 025004.

Under review

2025 J. J. Ziegler, K. Freese, **J. Lozano**, G. Montefalcone, *Explaining the "too massive" high-redshift galaxies in JWST data: numerical study of three effects and a simple relation* [arXiv:2507.21409]. In revision at MNRAS.

In preparation

2026 A. M. Alvertis and **J. Lozano-Mayo**, *Breakdown of exciton–phonon perturbation theory*. In preparation.

Collaboration papers

2024–2025 Member of the ALICE Collaboration. Papers carrying my name under ALICE's collective authorship policy are not listed individually.

Research Experience

2026–present **Excitons and phonons in strongly correlated materials**, *UT Austin*
With Antonios Alvertis. Computing exciton binding energies in strongly coupled electron-phonon systems with tensor-network methods, in the regime where perturbative treatments of the coupling break down.

2025–2026 **Generalized virial identities**, *UT Austin*, INDEPENDENT PROJECT
Derived a one-parameter family of virial identities for $O(n)$-symmetric solitons, instantons and bounces that resolves Derrick's global constraint into core and tail contributions. Verified them analytically for the Fubini-Lipatov instanton, the BPS monopole and the BPST instanton, and established their residuals as an accuracy diagnostic for numerical solutions.

2024–2025 **Charm production across color environments**, *ALICE Collaboration, UT Austin*
With Christina Markert. Measured h–D^0 azimuthal correlations in ALICE Run-3 Pb–Pb data to probe charm production in different color environments, and contributed performance studies of the upgraded Inner Tracking System (ITS2) as collaboration service work.

2024 **High-redshift galaxies and top-heavy initial mass functions**, *UT Austin*
With Katherine Freese. Modeled high-redshift galaxy spectral energy distributions with the population synthesis code PÉGASE to determine where JWST observations remain consistent with ΛCDM. UNDER REVIEW

2020–2022 **Solitons in scalar field theories with non-degenerate vacua**, *UNAM*
With Manuel Torres Labansat. Established the existence of static n-kink structures in deformed potentials by asymptotic analysis, characterized the inter-kink binding forces numerically, and derived an analytical relation for the multi-kink energy. Renormalized the one-loop quantum mass correction of kinks in a generalized ϕ^4 potential. TWO REFEREED PAPERS

Awards & Distinctions

2022 **Travel Grant**, German Physical Society (DPG) and International Association of Physics Students (IAPS), to compete in the PLANCKS international final at LMU Munich.

2021–2022 **Two-time Winner, Mexican Tournament of Physics** (*Torneo Mexicano de Física*), the national theoretical physics competition for bachelor's and master's students that serves as the Mexican PLANCKS preliminary. **1st of 30 teams** in 2022; the 2021 win placed me on the first Mexican team to compete at PLANCKS.

2021 **Mainz Physics Academy, PRISMA$^+$ Excellence Track Fellowship**, Johannes Gutenberg University Mainz: two-year M.Sc. scholarship (declined).

2021 **PLANCKS International Final, Porto**: **7th of 50 teams** in Mexico's first appearance at the international theoretical physics competition.

2021 **Juan Manuel Lozano Mejía Diploma**, UNAM Institute of Physics: institute-wide distinction for the most outstanding undergraduate thesis research in physics.

2021 **Honorific Mention**, B.Sc. thesis defense, UNAM: the highest distinction a thesis committee can award.

2019 **Research Assistantship**, UNAM Institute of Physics.

2017–2018 **Summer research and school scholarships**: Mexican Academy of Sciences Research Summer (2018); UNAM Experimental Physics School (2018); UNAM Physics School (2017).

Presentations

Talks at UT Austin

Apr 2026 *Exploration of electron-phonon interactions in strongly correlated materials*. Qualifying examination, Department of Physics.

Contributed talks

2025 *h–D^0 angular correlations and ΔROF performance studies at ITS2*. ALICE-USA Collaboration Meeting.

2021 *Multi-solitones en teorías escalares de campo con vacíos no-degenerados: el modelo doble de sine-Gordon*. LXVI National Physics Congress, Mexican Physical Society.

Teaching

2023–2026 **Teaching Assistant**, *The University of Texas at Austin*
Graduate Quantum Mechanics (2026); undergraduate Waves and Optics (2025), Modern Physics (2024–2025) and Electromagnetism Laboratory, PHY 105N (2023). Led discussion and laboratory sections, gave short pre-lab lectures, graded coursework.

2025 **Lecturer, Quantum Field Theory I**, *International Centre for Theoretical Physics (ICTP), Physics Without Frontiers*
Introductory quantum field theory course for the ICTP initiative in Latin America.

2024 **Teaching Assistant, Advanced Particle Physics**, *International Centre for Theoretical Physics (ICTP), Physics Without Frontiers*
Graduate course on advanced quantum field theory; led problem-set discussions.

2020–2022 **Teaching Assistant**, *UNAM, Faculty of Sciences*
Undergraduate Thermodynamics (2022) and Nuclear and Subnuclear Physics (2020). Ran problem-solving sessions; wrote and graded homework and exams.

Service & Outreach

2022 Referee, *Journal of Physics G: Nuclear and Particle Physics*.

2017 Volunteer instructor in physics and mathematics, UNAM university preparation course.

Skills, Training & Languages

Numerical Tensor networks (DMRG, TDVP), finite-difference PDE solvers

Programming Julia, Python, C++, Mathematica, ROOT/O^2 (ALICE), LaTeX

Schools CMB Data Analysis (UT Austin, 2025); Modeling and Tools for Data Analysis in Science (UANL, 2022); Bad Honnef Physics School on Effective and Lattice Field Theory (DPG, 2021); Mexican Schools on String Theory and Supersymmetry and on Astro-Particle Physics (Guanajuato, 2021)

Languages Spanish (native), English (C1), German (A2), French (A2)